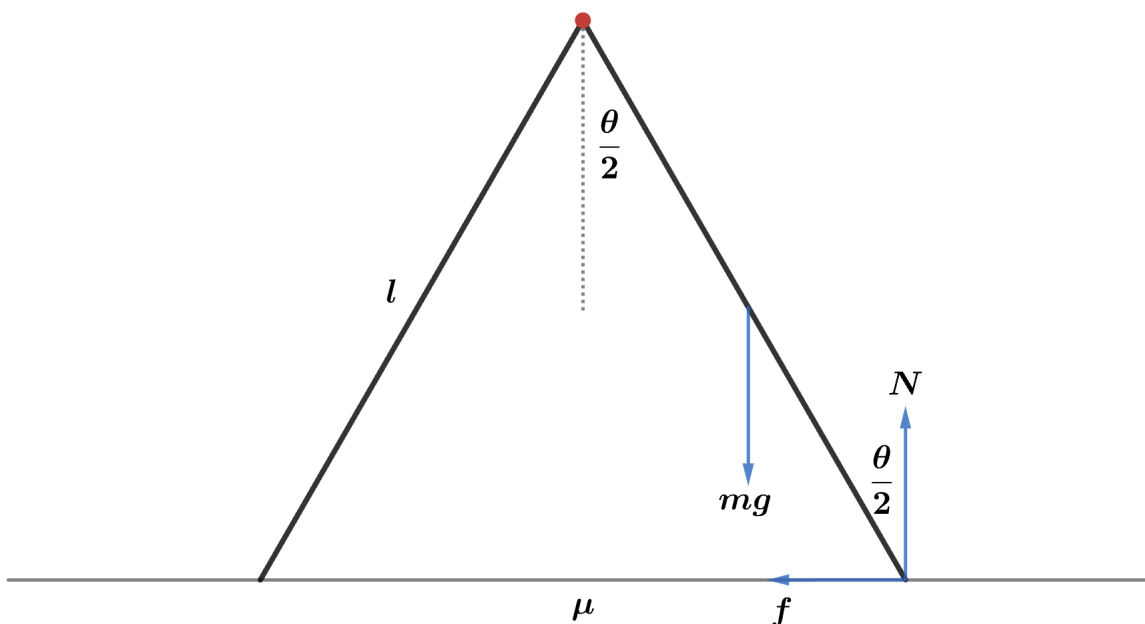


2013 F=ma Exam: Problem 4

Kevin S. Huang



We choose the hinge as the pivot point and balance torques on one of the legs. The weight mg of the leg acts at its center while the normal force $N = mg$ acts at the bottom to keep the leg at rest. Friction f points inward to prevent the sign from collapsing. These are the only forces that contribute to the torque since the force at the hinge has no moment arm. We have

$$mg \left(\frac{l}{2} \sin \frac{\theta}{2} \right) + f \left(l \cos \frac{\theta}{2} \right) = N \left(l \sin \frac{\theta}{2} \right) = mg \left(l \sin \frac{\theta}{2} \right)$$

Solving for f ,

$$\begin{aligned} fl \cos \frac{\theta}{2} &= mg \frac{l}{2} \sin \frac{\theta}{2} \\ f &= \frac{mg}{2} \tan \frac{\theta}{2} \end{aligned}$$

Since static friction $f \leq \mu N$, we have

$$\begin{aligned} \frac{mg}{2} \tan \frac{\theta}{2} &\leq \mu N = \mu mg \\ \tan \frac{\theta}{2} &\leq 2\mu \end{aligned}$$

so the answer is E.